Supreet Gulavani

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LinkedIn | GitHub
Location: Ratnagiri, Maharashtra, India

Professional Summary

Hardware Design & Verification Engineer with 1 yr 8 mo industry experience at HP, Intel, and NXP Semiconductors. Proficient in RTL design, FPGA bring-up, DMA engines, and PCIe. Skilled in embedded systems, simulation tools, and low-level hardware-software integration. Passionate about building high-performance digital systems, with research interests spanning Computer Architecture and Embedded Systems.

Education

Portland State University, Portland, OR, USA [3.36/4.00]
Master of Science in Electrical and Computer Engineering, Sep 2021 - Jun 2023

Savitribai Phule Pune University, Pune, MH, India [7.79/10.00]
Bachelor of Engineering in Electronics and Telecommunication, Aug 2016 - May 2020

Work Experience

HP (Contract), Firmware Engineer, Spring, TX, USA
May 2024 - Feb 2025
- Developed a DMA engine in Verilog on Lattice CertusPro-NX FPGA over AHBL via PCIe Gen3 X4
- Ported AHBL-based DMA engine to AXI-MM protocol, aligning with next-gen IP integration
- Collaborated on M.2 PCIe Gen3 debug tool design and documentation for software interfacing
- Integrated ADC IP to enable real-time temperature display
- Assisted in schematic reviews for tapeout
- Drafted testing scenarios for ARM-based NPU platforms, contributing to firmware feature development

Intel Corporation, Firmware (BIOS) Developer Intern, Hillsboro, OR, USA
Oct 2022 - Jun 2023
- Ported 20+ test cases to upcoming CPU platforms to enhance CI/CD coverage
- Refactored Slim Bootloader and resolved build script issues
- Authored whitepaper and open-source documentation to reduce internal-external discrepancies
- Created UEFI payload packages using Yocto and Windows platforms

NXP Semiconductors, DFT Intern, Austin, TX, USA
Jun 2022 - Sep 2022
- Built an automation tool for ATPG report analysis, improving test coverage assessment
- Worked on SCAN, MBIST, and IJTAG flows for subsystem-level regression

Barclays, Graduate Analyst, Pune, MH, India
Aug 2020 - Aug 2021
- Resolved API gateway issues and supported mission-critical systems, documenting all the processes

IoTIoT.in, AI Intern, Pune, MH, India
Dec 2018 - Feb 2019
- Built a CNN for object detection to assist with automated warehouse inventory

Technical Skills

Languages: SystemVerilog, Verilog, C, C++, Python, Bash
Tools: Vivado, ModelSim, Git, GDB, Make, Logic Analyzer, Oscilloscope
Hardware Architectures: RISC-V, x86, MIPS
Protocols: PCIe Gen3, AXI, AHB, Modbus, Wishbone, UART, SPI, I2C
Platforms: Windows, Linux, RTOS, FPGA (Xilinx A7, Lattice CertusPro), UEFI

Projects

LLM-based ICL/PDL Generator (Patent Pending)

Adiabatic Reversible Logic (In Progress) [GitHub]
- Conducting an independent (personal) study on low-power adiabatic reversible logic design techniques through academic papers
- Exploring the computing principles and implementing custom SystemVerilog simulations to test the energy-efficient logic gates

RISC-V 5-stage Pipelined CPU (In Progress)
- Designing a custom RISC-V CPU with a 5-stage pipeline architecture
- Implementing hazard detection and data forwarding units to resolve dependencies
- Simulating behavior in SystemVerilog using QuestaSim to verify correctness and performance
- Focusing on accuracy, throughput, and modular testbench development

MQTT-based IoT Weather Station - Nexys A7 FPGA
- Built an AXI peripheral-based Ethernet block and interfaced sensors via I2C
- Used LwIP stack with Microblaze CPU softcore to transmit data via MQTT on bare-metal firmware

Space Invaders on RISC-V Softcore - Nexys A7 FPGA [GitHub]
- Designed VGA Controller, push-button drivers in SystemVerilog, interfacing via Wishbone for RVFPGA CPU softcore
- Developed game logic and peripheral drivers in C

DDR4 DRAM Memory Controller [GitHub]
- Simulated command scheduling in C++ (in-order, FRFA, out-of-order) with queue logic and timing checks

Verification of RojoBlaze [GitHub]
- Wrote UVM-style testbench (BFM, scoreboard, monitor, driver) for the 8-bit microcontroller with a 4-stage pipeline
- Achieved 65% functional and 68% code coverage by feeding random constrained test cases through a Makefile

RISC-V ISA Simulator [GitHub]
- Built an RV32I simulator in C++ supporting single/multi-step breakpoints for debugging, MUL extensions, and syscall emulation

2D Multithreaded Graphics Renderer [Project Report]
- Implemented a multithreaded 2D renderer using HAGL SDL, studied FPS improvements across threads (1-16)

MLP-Aware Cache Replacement
- Simulated Qureshi et al.'s ISCA'06 (10.1109/ISCA.2006.5) cache replacement method

RL-based Channel Selection in Cognitive Radio Networks
- Used NS3 and OpenAI Gym for RL-based dynamic band/channel selection; reduced collisions over 75 episodes

Hand Gesture Recognition
- Built an IMU/flex-sensor-based glove that would detect the Indian Sign Language gestures
- Designed custom PCBs and programmed PIC & TIVA microcontrollers for communication and processing

Awards & Achievements

- **2nd Place**, Barclays Tech Innovation Challenge (2017- India)
- **3rd Place**, National Entrepreneur Challenge (2018 - India)